LEAD CITY UNIVERSITY
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<u>**Tutorial Questions**</u>

Question 1

a. Exhaustively, define what you understand by the term 'Computer system'?

Answer:

A **computer system** refers to an integrated set of hardware, software, and human interaction designed to perform data processing and information management tasks. Below is an exhaustive breakdown of what the term encompasses:

3marks

b. Discuss the following major hardware components of a computer system. Support your discussion with relevant diagrams.

a) Processor
b) Main memory
c) Secondary memory
d) Input devices
e) Output devices **2mark each**

Answer:

a) Processor (CPU) : The **processor** is the brain of the computer, responsible for executing instructions and performing calculations.

Diagram:

```
+-----------------+
|   CPU           |
| CU | ALU | Cache|
+-----------------+
```

b) Main Memory (Primary Memory): The computer's RAM holds data and instructions for immediate processing, while ROM stores essential startup instructions.

Diagram:

```
+------------------+
|  Main Memory     |
| RAM   |   ROM    |
+------------------+
```

c) Secondary Memory : Provides long-term, non-volatile storage for data and programs.

Diagram:

```
+---------------------+
|  Secondary Memory   |
| HDD | SSD | USB     |
+---------------------+
```

d) Input Devices : Capture data for the computer to process**.**

Diagram:

```
+------------------+
|  Input Devices   |
| Keyboard | Mouse |
+------------------+
```

e) **Output Devices :** Display or transmit processed data to users.

Diagram:

```
+------------------+
|  Output Devices  |
| Monitor | Printer|
+------------------+
```

B. What are the characteristics of secondary storage media? **2mark**

1. **Non-Volatile**: Retains data even when the power is off, ensuring long-term storage.
2. **High Capacity**: Can store large amounts of data, ranging from gigabytes to terabytes, suitable for extensive data needs.

Question 2

a. What are the differences between primary and secondary storage. **2.5marks**

Answer: .

- **Volatility**: Primary storage, like RAM, is volatile and loses data when the power is off, while secondary storage, like HDDs or SSDs, is non-volatile and retains data permanently.

- **Speed**: Primary storage is much faster and directly accessible by the CPU, whereas secondary storage is slower but designed for larger, long-term data storage

B Distinguish **2.5marks** between the following:

Answer:
(i) **Desktop Publishing Packages vs. Multimedia Packages**

- **Desktop Publishing**: These are tools used to design printed stuff like flyers, books, or posters. Examples: Adobe InDesign, Microsoft Publisher.
- **Multimedia**: These focus on creating or editing videos, music, and animations. Examples: Adobe Premiere, VLC Media Player.

(ii) **Input vs. Output**

- **Input**: Devices that send data into the computer, like a keyboard (typing) or a mouse (clicking).
- **Output**: Devices that show or play the results, like a monitor (shows visuals) or speakers (play sounds)

(iii) **Utility Software vs. Device Driver**

- **Utility Software**: Programs that keep your computer running smoothly, like antivirus software or disk cleanup tools.
- **Device Driver**: Special software that helps your computer communicate with hardware, like making your printer or graphics card work properly.

(iv) **Computer Hardware vs. Peripherals**

- **Hardware**: The core parts of your computer that are necessary to make it work, like the processor, RAM, or motherboard.
- **Peripherals**: The extras you connect to your computer to use it better, like keyboards, monitors, or printers.

b. Is a Graph plotter an Input device or an Output device? Describe how they work?
Answer:
A **graph plotter** is an **output device** that creates precise drawings like charts, maps, and blueprints on paper. It works by receiving data from a computer, processing it, and then using pens or ink to produce the final design. Commonly used for engineering and architectural plans

Question 3

a. Computers are classified according to:

a) Sizes **4marks**

Answer:
· **Supercomputers**: Extremely fast, used for simulations and research (e.g., IBM Summit).
· **Mainframes**: Handle massive data for organizations like banks (e.g., IBM Z-series).
· **Minicomputers**: Mid-range, used in businesses for database management.
· **Microcomputers**: Personal devices like laptops and desktops.

b) Types **3marks**
Answer:
· **Analog**: Process continuous data (e.g., thermometers).
· **Digital**: Process binary data, common in daily use (e.g., laptops).
· **Hybrid**: Combine analog and digital for specialized tasks (e.g., CT scanners).

c) Purposes, Discuss. **2marks**
Answer:
· **General-Purpose**: Versatile, for everyday tasks (e.g., laptops).
· **Special-Purpose**: Designed for specific jobs (e.g., ATMs).

b. Differentiate between the following terms:

(a) Data and Information.
(b) System software and application software.
(c) RAM and ROM
(d) Hardware and software
(e) Generic software and Custom software {**1mark each**}

Answer:

a) Data vs. Information

- **Data**: Raw, unprocessed facts (e.g., "25, 30").
- **Information**: Processed, meaningful data (e.g., "The average age is 27.5").

b) System Software vs. Application Software

- **System Software**: Runs the computer (e.g., Operating Systems like Windows).
- **Application Software**: Performs specific tasks (e.g., MS Word for writing).

c) RAM vs. ROM

- **RAM**: Temporary, fast memory for active tasks; loses data when off.
- **ROM**: Permanent memory that stores essential instructions like boot data.

d) Hardware vs. Software

- **Hardware**: Physical components (e.g., keyboard, CPU).
- **Software**: Programs that control hardware (e.g., browsers, games).

e) Generic vs. Custom Software

- **Generic Software**: Made for general use (e.g., MS Excel).
- **Custom Software**: Tailored for specific needs (e.g., a bank's transaction system).

Question 4

a. Write a short note on each of the following application software
   (i) Word Processor (ii). Spreadsheet (iii) Graphic Packages (iv). Database (v) Packages **2marks each**

b. Itemize 2 components of a Central Processing Unit and describe each of them in details. **5marks**

Answer:

· **Arithmetic Logic Unit (ALU)**:

- Performs all arithmetic operations (like addition, subtraction) and logical operations (like comparisons).
- Example: Deciding if one number is greater than another or calculating totals.

· **Control Unit (CU)**:

- Directs and coordinates the activities of all parts of the computer. It fetches instructions from memory and ensures they are executed properly.
- Example: Telling the ALU what to process and managing data flow between hardware components.

Question 5

a. Write a short note on the following; **15marks**
   i) First generation of computer
   ii) Second generation of computer
   iii) Third Generation of computer
   iv) Fourth Generation
   v) Modern computer

Answer:

· **First Generation (1940-1956)**:
These computers used **vacuum tubes** for processing and storage, making them large, slow, and prone to overheating. They consumed a lot of power and were difficult to maintain.

- **Example**: ENIAC, one of the earliest computers, filled an entire room and performed basic calculations.

· **Second Generation (1956-1963)**:
Transistors replaced vacuum tubes, which made computers smaller, faster, more reliable, and energy-efficient. These systems were still large but far more practical for businesses and scientific work.

- **Example**: IBM 1401, widely used for data processing tasks in businesses.

· **Third Generation (1964-1971)**:
The introduction of **integrated circuits** (ICs), which combined multiple transistors on a single chip, made computers even smaller, faster, and more affordable. These computers could support a wider range of applications.

- **Example**: IBM System/360, which supported different applications and was highly adaptable.

· **Fourth Generation (1971-Present)**:
Microprocessors, which are tiny chips containing millions of transistors, revolutionized computing. This era saw the rise of personal computers, advanced software, and more user-friendly interfaces.

- **Example**: Early personal computers like the Apple II and modern devices like laptops and desktops.

· **Modern Computers**:
Today's computers are powerful, compact, and portable. They include **smartphones**, **tablets**, and **cloud computing** devices, which use advanced processors and AI for tasks ranging from gaming to complex problem-solving.

- **Example**: Devices like the MacBook Pro, Google Pixel, and advanced AI-powered systems.

Question 6

a. Discuss the application of computer system in any 3 areas of your choice. **2marks each**

Answer:

· **Healthcare**:
Computers assist in disease diagnosis, storing patient records, and running medical imaging systems (e.g., MRI, CT scans).

· **Education**:
Computers support online learning, virtual classrooms, and interactive educational software.

· **Banking and Finance**:
Computers handle online banking, transaction processing, and account management securely.

b. With the aid of specific examples, discuss the role of computer to modern day era,

Answer:

· **Communication**: Email, social media, and video calls (e.g., Zoom) are possible thanks to computers.

· **Business**: Companies use computers for managing operations, marketing, and customer service, leading to faster decision-making and greater efficiency (e.g., Amazon's online store).

· **Entertainment**: Computers power video games, streaming services (like Netflix), and digital content creation (e.g., YouTube).

c. Distinguish between (1) RAM and ROM (2) input and output unit (3) primary and auxiliary memory. **2marks each.**

Answer:

1. RAM (Random Access Memory): is volatile memory, meaning it loses data when power is off. It temporarily holds data and instructions needed for processing tasks. For example, it runs applications while the computer is on.

ROM (Read-Only Memory) : on the other hand, is non-volatile and retains data permanently. It stores essential instructions like the computer's startup processes (BIOS).

2.The input unit sends data into the computer for processing, using devices like keyboards or mice.
The output unit displays or conveys processed information to the user through devices such as monitors, printers, or speakers.

3.Primary memory is fast and volatile, used for temporary data storage during processing. Examples include RAM and ROM.
Auxiliary memory is slower but non-volatile, designed for long-term storage of data. Examples include hard drives, SSDs, and USB drives.

Question 7

a. There are several ways that signed numbers can be represented in binary, but the most common representation used today is called two's complement. (i) Briefly explain two different ways in which it can be used, and (ii) why it is used. **3marks**

Answer:

(i) Two Ways Two's Complement Can Be Used

Representing Positive Numbers:

1. Positive numbers are written in regular binary form. For example, +5 in 8-bit binary is simply 00000101, just like in normal binary notation.

Representing Negative Numbers:

For negative numbers, the process involves:

1. Writing the binary equivalent of the positive version of the number (e.g., +5 → 00000101).
2. Inverting all the bits (flip 0s to 1s and 1s to 0s), turning 00000101 into 11111010.
3. Adding 1 to the result: 11111010 + 1 = 11111011.

So, -5 is represented as 11111011 in two's complement.

(ii) Why Two's Complement is Used

Simplified Arithmetic:

Two's complement makes addition and subtraction of signed numbers simpler because it allows the same arithmetic operations to be used for both positive and negative numbers. No special handling is needed for negative numbers.

Unique Zero Representation:

Two's complement ensures that there is only one representation for zero (00000000 for 8-bit), avoiding complications that arise in other systems like sign-magnitude, which have both +0 and -0.

Efficient Hardware Implementation:

It's easier and faster to implement in hardware since the system doesn't need to handle separate sign bits. The entire number is treated as a binary value, simplifying the design of processors and circuits.

a. Briefly discuss four divisions ofNumber Systems **4marks**

Answer:

· **Binary (Base-2)**:

- Uses only two digits: 0 and 1.
- Commonly used in computers because they operate on electrical signals (on/off states).
- Example: 1010 in binary = 10 in decimal.

b. Highlight 3 reasons why hexadecimal numbers are used. **3marks**

Answer:

· **Compact Representation**:
Hexadecimal is more compact than binary, as one hex digit represents four binary digits, making large binary numbers easier to read.

· **Ease of Conversion**:
Converting between binary and hexadecimal is simple since every four binary digits map directly to one hex digit.

· **Memory and Address Representation**:
Hexadecimal is commonly used for memory addresses and machine code, as it's easier for humans to read than long binary strings.

c. Obtain the 1's and 2's complements of the following 8-bit numbers: i) 10101110 ii) 10000000 iii) 10000001 iv)**3marks**

Answer:

To compute the **1's complement** and **2's complement** for a given binary number:

1's Complement: Flip every bit (change 0s to 1s and 1s to 0s).

2's Complement: Take the 1's complement and **add 1** to the result.

Now, let's compute step-by-step for each number:

i) 10101110

1's Complement: Invert bits → 01010001

2's Complement: Add 1 → 01010001 + 1 = 01010010

ii) 10000000

1's Complement: Invert bits → 01111111

2's Complement: Add 1 → 01111111 + 1 = 10000000

iii) 10000001

1's Complement: Invert bits → 01111110

2's Complement: Add 1 → 01111110 + 1 = 01111111

Question 8

a. Perform the arithmetic operations (+42) + (-13) and 12810 − 6410 in binary using 2's complement representation for negative numbers **5marks**

Answer:

(+42) + (-13)

Convert Numbers to Binary:

1. +42 → 00101010 (binary for 42).
2. -13 → First, write +13 in binary (00001101), flip the bits (11110010), and add 1 → 11110011 (this is -13 in 2's complement).

00101010 (42)

+ 11110011 (-13)

00011101 (+29 in binary)

(128 - 64)

Convert Numbers to Binary:

1. +128 → 10000000 (binary for 128).
2. -64 → Write +64 in binary (01000000), flip the bits (10111111), and add 1 → 11000000 (this is -64 in 2's complement).

10000000 (128)

+ 11000000 (-64)

01000000 (+64 in binary)

Decimal (Base-10):

- The standard system for humans, using ten digits: 0 to 9.
- Each digit's position represents powers of 10.
- Example: 345 = 3×102+4×101+5×100=3453 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 = 3453×102+4×101+5×100=34

Octal (Base-8):

- Uses eight digits: 0 to 7.
- Often used in computing as a shorthand for binary (each octal digit represents three binary bits).
- Example: 12 in octal = 1×81+2×80=101 \times 8^1 + 2 \times 8^0 = 101×81+2×80=10 in decimal.

Hexadecimal (Base-16):

- Uses sixteen symbols: 0 to 9 and A to F (where A=10, B=11, etc.).
- Common in computing for representing large binary numbers concisely (each hex digit represents four binary bits).
- Example: 1F in hexadecimal = 1×161+15×160=311 \times 16^1 + 15 \times 16^0 = 311×161+15×160=31 in decimal.

c. Perform the subtraction with the following unsigned binary numbers by taking the 2's complement of the subtrahend i) 11010 – 10000 ii) 11010 – 1101 iii) 100 – 110000 iv) 1010100 – 1010100 **6marks**

Answer:

i) 11010 - 10000

Write the Numbers in Binary:

1. Minuend = 11010
2. Subtrahend = 10000

Find the 2's Complement of the Subtrahend:

1. Flip the bits of 10000: 01111
2. Add 1: 01111 + 1 = 10000

11010

+10000

01010 (No carry-over, result = 1010 in binary or 10 in decimal)

ii) 11010 - 1101

11010

+10011

10101 (Result = 10101 in binary or 21 in decimal)

iii) 100 - 110000

000100

+010000

010100 (Result = 10100 in binary or 20 in decimal)

iv) 1010100 - 1010100

1010100

+0101100

0000000 (Result = 0 in binary or 0 in decimal)

Question 9
In detailed format, describe the following computer terms and give appropriate examples:
(i) Utility (ii) Interpreter (iii) Editor (iv) Assembler (v) Compiler
3 marks each = 15 marks

Answer:

(i) Utility:
A **utility** is a type of system software designed to help manage, maintain, and optimize computer resources and operations. Utilities perform specific tasks to improve system performance or fix issues.

- **Examples**:
    1. **Disk Cleanup**: Removes unnecessary files to free up storage.
    2. **Antivirus Software**: Protects the system from malware.

(ii) **Interpreter:**

An **interpreter** is a program that translates high-level programming code into machine code **line by line** and executes it immediately. Unlike a compiler, an interpreter does not generate a separate executable file; it executes code on the fly.

Examples:

1. **Python Interpreter**: Executes Python scripts.
2. **JavaScript Interpreter**: Runs JavaScript code in browsers.

(iii) Editor

An **editor** is a software tool used to create, edit, or modify text or code. It is an essential tool for developers, writers, and data analysts. Editors often include features like syntax highlighting, search-and-replace, and auto-completion to enhance productivity.

- **Examples**:
    1. **Notepad**: A basic text editor for simple documents.
    2. **Visual Studio Code**: A feature-rich editor for coding in multiple languages.

(iv) Assembler

An **assembler** is a program that converts assembly language code into machine code (binary instructions that a CPU can execute). Assembly language is a low-level programming language that uses symbolic instructions and is specific to a processor.

Example:

- Writing assembly code for Intel processors and using **MASM (Microsoft Macro Assembler)** to generate machine code.

(v) Compiler

A **compiler** is a program that translates high-level programming code (like C, Java, or C++) into machine code all at once, producing an executable file. Unlike an interpreter, a compiler processes the entire program before execution.

Examples:

1. **GCC (GNU Compiler Collection)**: Compiles C/C++ programs.
2. **Javac**: The Java compiler used to create .class files from .java source files.

Question 10
Write short notes on each of the following with appropriate examples
Word Processor (ii) Spreadsheet (iii) Presentation software (iv) Graphics software (v) PHP **3 marks each = 15 marks**

Answer:

(i) Word Processor

A **word processor** is a software application used for creating, editing, formatting, and printing text documents. It includes tools for spell-checking, grammar correction, and text styling.

Examples:

1. **Microsoft Word**: Used for professional document creation.

(ii) Spreadsheet

A **spreadsheet** is software designed for organizing, analyzing, and storing data in tabular form. It supports functions for calculations, data visualization (charts/graphs), and financial modeling.

Examples:

1. **Microsoft Excel**: Advanced features like pivot tables and formulas

(iii) Presentation Software

Presentation software is used to create visual slideshows that include text, images, charts, animations, and multimedia elements to communicate ideas effectively.

Examples:

1. **Microsoft PowerPoint**: Widely used for business and educational presentations.

(iv) Graphics Software

Graphics software is used for creating and editing visual content like images, designs, and animations. It provides tools for drawing, photo editing, and vector graphics creation.

Examples:

1. **Adobe Photoshop**: Used for image editing and graphic design.

(v) PHP (Hypertext Preprocessor)

PHP is a server-side scripting language primarily used for web development. It enables the creation of dynamic and interactive web pages.

Examples:

1. **WordPress**: Built using PHP for content management systems.

Question 11

a. Describe the strengths and limitations of artificial intelligence. **7marks**

Answer:

Strengths	Limitations		
Automation: AI automates repetitive tasks, improving efficiency and accuracy.	**High Development Costs**: Building and maintaining AI systems is expensive.		
Data Analysis: Processes large datasets quickly to identify trends and patterns.	**Data Dependency**: Requires vast amounts of high-quality data to perform effectively.		
Personalization: Provides tailored recommendations and services.	**Lack of Creativity**: Cannot think outside predefined algorithms or innovate.		
24/7 Operation: Works continuously without breaks, unlike humans.	**Ethical Concerns**: May perpetuate biases in training data or be misused.		
Problem-Solving: Excels in solving complex problems in fields like medicine.	**Limited Understanding**: Lacks human intuition and contextual awareness.		
Improved Accuracy: Enhances precision in decision-making.	**Job Displacement**: Automation can replace human jobs in some industries.		
Adaptability: Learns and improves over time through machine learning.	**Vulnerability**: Susceptible to cyberattacks or technical malfunctions.		

b. Discuss the social impact of artificial intelligence to the society. **8marks**

Answer:

Positive Impacts	Negative Impacts		
Improved quality of life with automation	Job displacement due to automation		
Advances in healthcare (e.g., AI diagnostics)	Privacy invasion through surveillance		
Education accessibility via personalized learning	Bias and discrimination in AI decisions		
Economic growth through innovation	Social isolation from reduced human interaction		

Question 12

a. The computer software could be divided into two major groups namely System Software and Application Software. Briefly explain each with examples. **9marks**

Answer:

1. System Software

System software is designed to manage and control the computer hardware and provide a platform for running application software. It acts as an intermediary between hardware and user applications.

Key Functions:

- Manages hardware resources like CPU, memory, and storage.
- Provides a user interface (e.g., command line or graphical).

Examples:

- **Operating Systems (OS)**: Windows, macOS, Linux.
- **Utility Software**: Antivirus programs, disk management tools.

2. Application Software

Application software is developed for end-users to perform specific tasks or solve problems. These programs run on top of system software and cater to individual or business needs.

Key Functions:

- Allows users to create, edit, and manage content.
- Enhances productivity and functionality for specific purposes.

Examples:

- **Productivity Software**: Microsoft Word (word processing), Excel (spreadsheets).
- **Media Players**: VLC Media Player, Windows Media Player.

b. Define the terms:

i. Programming (program Design) ii. Algorithm iii. Pseudocode. **6marks**

Answer:

i. Programming (Program Design)

Programming is the process of creating instructions (a program) that a computer can execute to perform a specific task. Program design involves planning and organizing the program's structure and logic before actual coding begins.

ii. Algorithm

An algorithm is a step-by-step procedure or set of rules to solve a problem or perform a task. It is written in a logical order and serves as the foundation for programming

iii. Pseudocode

Pseudocode is a simplified, human-readable representation of an algorithm that combines programming logic with natural language. It is used to plan and visualize program structure before coding.

Question 13

a. Discuss the following in relation to Cyber ethics:

(i) Privacy (ii) Security and crime (iii) Free expression and content control **9marks**

Answer:

i. Privacy

Privacy in the digital age refers to the right of individuals to control their personal information and decide how it is collected, used, or shared online.

Example: Social media platforms collecting user data to show targeted ads raise privacy concerns if done without consent.

ii. Security and Crime

Cybersecurity involves protecting digital systems, networks, and data from unauthorized access, damage, or theft. Cybercrime includes illegal activities conducted online, such as hacking, phishing, or identity theft.

Example: Ransomware attacks demand payment from victims to restore access to their data, violating ethical and legal standards

iii. Free Expression and Content Control

Free expression online refers to the ability to share opinions and ideas freely, while content control involves regulating online content to prevent harm or illegal activities.

> **Example**: Content moderation on platforms like Twitter ensures harmful content is removed but may raise concerns about limiting free expression.

b. Explain the meaning of the term "*Cyber ethics*" and its two approaches **6marks**

Answer:

Cyber ethics refers to the moral principles and guidelines that govern behavior in the digital world. It involves understanding and practicing responsible use of technology, addressing issues such as privacy, security, intellectual property, and online conduct. Cyber ethics aims to ensure fairness, respect, and accountability in digital interactions.

Deontological Approach

1. Focuses on rules and duties that dictate ethical behavior in the cyber world.
2. Actions are judged as right or wrong based on adherence to these rules, regardless of the consequences.

Example: Always obtaining user consent before collecting personal data, as it is a moral obligation.

Consequentialist (Utilitarian) Approach

1. Evaluates actions based on their outcomes or consequences.
2. An action is considered ethical if it produces the greatest good for the greatest number of people.

Example: Blocking harmful content on social media to protect public safety, even if it limits some individuals' free speech.

Question 14

a. Explain the following terms:

(i). Machine language (ii).High **level** language (iii). Assembly and symbolic language. **9marks**

Answer:

i. Machine Language

Machine language is the lowest-level programming language that a computer can directly understand and execute. It consists of binary code (0s and 1s) specific to a computer's hardware.

> **Example**: A machine instruction in binary: 11001010.

ii. High-Level Language

High-level languages are programming languages designed to be human-readable and easier to write. These languages use syntax and constructs similar to natural languages, abstracting away hardware complexities.

> **Examples**: Python, Java, C++, and JavaScript.

iii. Assembly and Symbolic Language

Assembly language is a low-level language that uses symbolic instructions to represent machine code. Each assembly instruction corresponds to a specific machine instruction.

Example:
Machine code: 1011 0000 0110.
Assembly language: MOV AX, 06.

b. Simplify the following i. 1100 x 101 ii. 10101 x 111 iii. 1111 + 111 + 1111 iv. 101011 – 10010 **6marks**

Answer:

```
   i. 1100 × 101
    1100    (12 in decimal)
  ×  101     (5 in decimal)
  --------
    1100    (1100 × 1)
   0000     (1100 × 0, shifted one position to the left)
+ 1100      (1100 × 1, shifted two positions to the left)
  --------
  111100    (Result: 60 in decimal)
```

```
   ii. 10101 × 111
       10101    (21 in decimal)
     ×   111     (7 in decimal)
     --------
       10101    (10101 × 1)
      10101     (10101 × 1, shifted one position to the left)
   + 10101      (10101 × 1, shifted two positions to the left)
     --------
     1111101    (Result: 147 in decimal)
```

iii. 1111 + 111 + 1111

```
  1111
+  111
+ 1111
--------
 10111   (Result: 23 in decimal)
```

iv. 101011 - 10010

```
 101011   (43 in decimal)
-  10010   (18 in decimal)
 --------
  11101   (Result: 25 in decimal)
```

Question 15

a. Convert the following Hexadecimal numbers to binary.

i) $3B29_{16}$ ii) $9C32_{16}$ iii) $0A2B_{16}$ **6marks**

Answer:

i. $3B29_{16}$ (Hexadecimal to Binary)

1. Convert each hexadecimal digit to its 4-bit binary equivalent:
   1. **3 → 0011**
   2. **B → 1011**
   3. **2 → 0010**
   4. **9 → 1001**

So, **$3B29_{16}$** in binary is:
0011 1011 0010 1001

ii. $9C32_{16}$ (Hexadecimal to Binary)

1. Convert each hexadecimal digit to its 4-bit binary equivalent:
   1. **9 → 1001**
   2. **C → 1100**
   3. **3 → 0011**
   4. **2 → 0010**

So, **$9C32_{16}$** in binary is:
1001 1100 0011 0010

iii. $0A2B_{16}$ (Hexadecimal to Binary)

1. Convert each hexadecimal digit to its 4-bit binary equivalent:
    1. **0 → 0000**
    2. **A → 1010**
    3. **2 → 0010**
    4. **B → 1011**

So, **$0A2B_{16}$** in binary is:
0000 1010 0010 1011

b. State 3 advantages of hexadecimal number system over binary number system. **6marks**

Answer:

Advantage	Why It's Helpful
Shorter and Simpler	Hexadecimal takes fewer digits to represent the same value as binary.
Easier to Convert	Converting between binary and hexadecimal is quick and direct.
Easier to Read	Hexadecimal is easier to understand than long binary numbers.

c. Simplify: i. 1111 x 10 ii. 10110 x 110 iii. 10101 x 101 **3marks**

Answer:

i. 1111 × 10
So, 1111 × 10 = 11110.

ii. 10110 × 110
So, 10110 × 110 = 1111000.

iii. 10101 × 101
So, 10101 × 101 = 1101101.

……………………………………………………..

HOD Sign